

Arithmetic Rademacher Transseries for the Partition Numbers

Exact Exponential Sectors, All-Order Coefficients, and Asymptotic Inversion

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Abstract

Let $p(n)$ be the ordinary partition number. The familiar Hardy–Ramanujan term captures only the first level of a much richer asymptotic object. Starting from Rademacher’s convergent series, we introduce the shifted phase

$$N = n - \frac{1}{24}, \quad x = \pi\sqrt{\frac{2N}{3}}, \quad C = \frac{\pi^2}{6\sqrt{3}},$$

and prove the exact sector decomposition

$$p(n) = \frac{C}{x^2} \sum_{k \geq 1} \frac{A_k(n)}{\sqrt{k}} \left\{ e^{x/k} \left(1 - \frac{k}{x} \right) + e^{-x/k} \left(1 + \frac{k}{x} \right) \right\}.$$

Thus the full forward transseries is not guessed from a divergent power series: it is the reorganized, convergent Rademacher expansion itself. Each sector has only a linear amplitude in x^{-1} in the natural shifted variable. Re-expansion in $n^{-1/2}$ produces an infinite coefficient sequence. We give (i) a coefficient-generating function valid for every k and both signs, (ii) a complete Bell-polynomial formula and a triangular recurrence for all coefficients, and (iii) a proof of the finite O’Sullivan formula for the dominant coefficients by a Catalan change of variables and Lagrange inversion.

The inverse problem has two layers. The elementary carrier $y = Ce^X/X^2$ is inverted exactly by

$$X = -2W_{-1} \left(-\frac{1}{2} \sqrt{\frac{C}{y}} \right).$$

Its complete iterated-logarithm expansion has coefficient polynomials

$$P_j(L) = 2^{j+1} \sum_{q=1}^j \frac{s(j, q)}{(j+1-q)!} L^{j+1-q}, \quad L = \log \log(y/C),$$

where $s(j, q)$ are signed Stirling numbers of the first kind. Inverting the full dominant Rademacher sector adds a convergent rational series $U(1/X) = \sum b_m X^{-m}$; we derive a coefficient recurrence and list its first terms. Finally, because $A_k(n)$ is periodic modulo k , the exponentially small inverse sectors form an arithmetic family. At every finite Rademacher level K , fixing n modulo $\text{lcm}(1, \dots, K)$ turns the problem into an ordinary analytic transseries inversion. We give both a multivariate Newton–Hensel recurrence for every sector coefficient and a closed Lagrange–Buermann/Bell-polynomial formula. Compatible residue classes yield a natural profinite phase parameter in $\widehat{\mathbb{Z}}$.

We distinguish this smooth/asymptotic inverse from the integer-valued generalized inverse of the discrete sequence. A final rounding layer is unavoidable for arbitrary real ordinates, while at exact ordinates $y = p(n)$ the dominant-sector inverse already differs from n by an exponentially small quantity and therefore recovers n by nearest-integer rounding for all sufficiently large n . For completeness, we also give the full transseries of the multiplicative reciprocal $1/p(n)$.

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1 Introduction and statement of the program

The partition function $p(n)$ is defined by

$$\sum_{n \geq 0} p(n) q^n = \prod_{m \geq 1} \frac{1}{1 - q^m}, \quad |q| < 1. \quad (1.1)$$

It is one of the central coefficient sequences in additive number theory. Hardy and Ramanujan proved the celebrated leading approximation

$$p(n) \sim \frac{1}{4\sqrt{3}n} \exp\left(\pi\sqrt{\frac{2n}{3}}\right), \quad (1.2)$$

and Rademacher replaced the asymptotic formula by an exact convergent series involving Dedekind sums. Modern work gives all algebraic orders and explicit error bounds; see Hardy–Ramanujan [7], Rademacher [16, 17, 18], Lehmer [9, 10, 11], O’Sullivan [13], and Banerjee–Paule–Radu–Schneider [2]. The exact Rademacher formula and its leading asymptotic were also formalized in Isabelle/HOL in 2026 [6].

The purpose of this article is not to reprove the circle method. We take Rademacher’s theorem as the analytic input and develop, from it, a complete coefficient calculus for the forward and inverse transseries. This choice has three advantages.

1. The complete exponential hierarchy is present from the beginning. There is no need to infer exponentially small sectors from a divergent perturbative series.
2. The natural shifted phase $x = \pi\sqrt{2(n-1/24)}/3$ turns every Rademacher summand into two exponentials with elementary amplitudes.
3. All inverse coefficients can be generated by triangular recurrences, Bell polynomials, and Lagrange–Bürmann formulas of the kind developed in the accompanying coefficient-calculus and transseries documents of the ProveIt repository [14, 15].

There is an important semantic issue. The phrase “inverse of the partition numbers” can mean either the reciprocal $1/p(n)$ or the inverse function that recovers n from a large value of $p(n)$. The main subject of this article is the latter. Because p is a sequence, however, an inverse function is not canonical until one chooses an interpolation or restricts to the exact range $\{p(n)\}$. We make this distinction precise in [section 8](#). The multiplicative reciprocal is simpler and is treated in [section 7](#).

1.1 The main formulas at a glance

Set

$$a = \frac{1}{24}, \quad N = n - a, \quad \lambda = \pi\sqrt{\frac{2}{3}}, \quad x = \lambda\sqrt{N}, \quad C = \frac{\pi^2}{6\sqrt{3}}. \quad (1.3)$$

Then the exact forward transseries is

$$p(n) = \frac{C}{x^2} \sum_{k \geq 1} \frac{A_k(n)}{\sqrt{k}} \left[e^{x/k} \left(1 - \frac{k}{x}\right) + e^{-x/k} \left(1 + \frac{k}{x}\right) \right]. \quad (1.4)$$

In unshifted powers $t = n^{-1/2}$, the (k, σ) sector, $\sigma \in \{+1, -1\}$, is

$$\frac{A_k(n)}{4\sqrt{3k}n} e^{\sigma\lambda\sqrt{n}/k} \sum_{m \geq 0} c_{k,\sigma,m} n^{-m/2}, \quad (1.5)$$

where

$$\sum_{m \geq 0} c_{k,\sigma,m} t^m = (1 - at^2)^{-1} \exp \left[\frac{\sigma\lambda}{kt} \left(\sqrt{1 - at^2} - 1 \right) \right] \left(1 - \frac{\sigma kt}{\lambda\sqrt{1 - at^2}} \right). \quad (1.6)$$

This one formula generates every algebraic coefficient in every Rademacher sector.

For the inverse, let

$$R = \log \frac{y}{C}, \quad X = -2W_{-1} \left(-\frac{1}{2} \sqrt{\frac{C}{y}} \right). \quad (1.7)$$

Then $X - 2 \log X = R$. If $L = \log R$, the complete polynomial–logarithmic expansion is

$$X \sim R + 2L + \sum_{j \geq 1} \frac{P_j(L)}{R^j}, \quad P_j(L) = 2^{j+1} \sum_{q=1}^j \frac{s(j, q)}{(j+1-q)!} L^{j+1-q}. \quad (1.8)$$

The inverse of the full dominant positive sector is

$$x_0 = X + U(X^{-1}), \quad U(z) = \sum_{m \geq 1} b_m z^m, \quad (1.9)$$

where U is the unique formal solution of

$$U + \log(1 - z + zU) - 3 \log(1 + zU) = 0. \quad (1.10)$$

The complete arithmetic exponential correction is then obtained by either of two equivalent general formulas:

- the coefficient-by-coefficient recurrence in [theorem 11.1](#);
- the Lagrange–Bürmann composition formula in [theorem 11.3](#).

Finally,

$$n = a + \frac{x^2}{\lambda^2} = \frac{1}{24} + \frac{3x^2}{2\pi^2}. \quad (1.11)$$

2 Notation, Dedekind sums, and coefficient polynomials

2.1 Dedekind sums and Rademacher coefficients

For a real number u , define the sawtooth function

$$((u)) = \begin{cases} u - \lfloor u \rfloor - \frac{1}{2}, & u \notin \mathbb{Z}, \\ 0, & u \in \mathbb{Z}. \end{cases} \quad (2.1)$$

The Dedekind sum is

$$s(h, k) = \sum_{j=1}^{k-1} \left(\left(\frac{j}{k} \right) \right) \left(\left(\frac{hj}{k} \right) \right). \quad (2.2)$$

We use the real normalization

$$A_k(n) = \sum_{\substack{1 \leq h \leq k \\ (h, k)=1}} \cos \left(\pi \left(s(h, k) - \frac{2nh}{k} \right) \right). \quad (2.3)$$

The equivalent complex-exponential version may be used throughout. Two elementary properties will repeatedly matter:

$$A_k(n+k) = A_k(n), \quad |A_k(n)| \leq \varphi(k) \leq k. \quad (2.4)$$

The first follows because shifting n by k changes every cosine argument by an integral multiple of 2π ; the second is the triangle inequality. The first few amplitudes are

$$A_1(n) = 1, \quad A_2(n) = (-1)^n, \quad (2.5)$$

$$A_3(n) = 2 \cos \left(\frac{2\pi n}{3} - \frac{\pi}{18} \right), \quad A_4(n) = 2 \cos \left(\frac{\pi n}{2} - \frac{\pi}{8} \right). \quad (2.6)$$

The periodicity in [equation \(2.4\)](#) is the source of the arithmetic phase in the inverse transseries.

2.2 Bell and Stirling conventions

The exponential partial Bell polynomials $B_{m,q}$ are defined by

$$\frac{1}{q!} \left(\sum_{j \geq 1} x_j \frac{z^j}{j!} \right)^q = \sum_{m \geq q} B_{m,q}(x_1, \dots, x_{m-q+1}) \frac{z^m}{m!}. \quad (2.7)$$

The complete Bell polynomial is

$$Y_m(x_1, \dots, x_m) = \sum_{q=0}^m B_{m,q}(x_1, \dots, x_{m-q+1}). \quad (2.8)$$

Consequently, if $F(z) = \exp(\sum_{j \geq 1} \ell_j z^j)$, then

$$[z^m]F(z) = \frac{1}{m!} Y_m(1!\ell_1, 2!\ell_2, \dots, m!\ell_m). \quad (2.9)$$

This is precisely the exponential formula used in combinatorial coefficient calculus.

The signed Stirling numbers of the first kind $s(m, q)$ are fixed by

$$u(u-1) \cdots (u-m+1) = \sum_{q=0}^m s(m, q) u^q, \quad (2.10)$$

and equivalently

$$(\log(1+z))^q = q! \sum_{m \geq q} s(m, q) \frac{z^m}{m!}. \quad (2.11)$$

2.3 What “full transseries” means here

A single one-variable Hahn series is not the most faithful global model for the Rademacher expansion. The relative positive-sector exponents

$$1 - \frac{1}{k} \quad (k \geq 2)$$

increase to 1, while the reflected exponents

$$1 + \frac{1}{k} \quad (k \geq 2)$$

decrease to 1. In particular, the latter set is not well ordered in the usual asymptotic ordering. We therefore use the following two complementary notions.

Definition 2.1 (Exact analytic transseries). *The full forward transseries is the convergent Rademacher sum [equation \(1.4\)](#), regarded as an exact sum of exponential sectors.*

Definition 2.2 (Finite-level formal transseries). *For $K \geq 1$, retain the sectors with $1 \leq k \leq K$ and treat their distinct exponentials as independent formal variables. All coefficient extraction and inversion is then performed in an ordinary finitely generated multivariate power-series ring. The limit $K \rightarrow \infty$ is taken analytically through Rademacher convergence, not by pretending that the entire support is a locally finite one-variable Hahn grid.*

This finite-level viewpoint is more than a technical convenience: it is what makes the periodic amplitudes $A_k(n)$ compatible with inversion.

3 Rademacher's formula in exact transseries normal form

We take as input Rademacher's convergent formula

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k \geq 1} A_k(n) \sqrt{k} \frac{d}{dn} \left(\frac{\sinh(x/k)}{\sqrt{N}} \right), \quad (3.1)$$

with the variables in [equation \(1.3\)](#). This normalization agrees with the modern formalization in [6].

Theorem 3.1 (Exact elementary sector decomposition). *For every integer $n \geq 1$,*

$$p(n) = \frac{C}{x^2} \sum_{k=1}^{\infty} \frac{A_k(n)}{\sqrt{k}} \left\{ e^{x/k} \left(1 - \frac{k}{x} \right) + e^{-x/k} \left(1 + \frac{k}{x} \right) \right\}. \quad (3.2)$$

The series is convergent and has exactly the same sum as Rademacher's series.

Proof. Since $dx/dn = x/(2N)$,

$$\frac{d}{dn} \left(N^{-1/2} \sinh(x/k) \right) = \frac{1}{2N^{3/2}} \left(\frac{x}{k} \cosh(x/k) - \sinh(x/k) \right) \quad (3.3)$$

$$= \frac{x}{4kN^{3/2}} \left\{ e^{x/k} \left(1 - \frac{k}{x} \right) + e^{-x/k} \left(1 + \frac{k}{x} \right) \right\}. \quad (3.4)$$

Multiplying by $A_k(n)\sqrt{k}/(\pi\sqrt{2})$ gives

$$\frac{A_k(n)}{4\sqrt{3k}N} \left\{ e^{x/k} \left(1 - \frac{k}{x} \right) + e^{-x/k} \left(1 + \frac{k}{x} \right) \right\},$$

because $x/(\pi\sqrt{2}\sqrt{N}) = 1/\sqrt{3}$. Finally,

$$\frac{1}{4\sqrt{3k}N} = \frac{C}{\sqrt{k}x^2}$$

by $x^2 = (2\pi^2/3)N$. The transformation is termwise algebra, so convergence is inherited from [equation \(3.1\)](#). $\square$

Corollary 3.2 (Dominant sector and exact relative correction). *Define*

$$P_+(x) = C e^x x^{-2} \left(1 - \frac{1}{x} \right). \quad (3.5)$$

Then

$$p(n) = P_+(x)(1 + \mathcal{E}(x;n)), \quad (3.6)$$

where

$$\begin{aligned} \mathcal{E}(x;n) = & e^{-2x} \frac{1+x^{-1}}{1-x^{-1}} \\ & + \sum_{k \geq 2} \frac{A_k(n)}{\sqrt{k}} \left\{ e^{-(1-1/k)x} \frac{1-k/x}{1-1/x} + e^{-(1+1/k)x} \frac{1+k/x}{1-1/x} \right\}. \end{aligned} \quad (3.7)$$

Proof. Separate the $k = 1$ positive exponential in [equation \(3.2\)](#) and divide all remaining terms by it. $\square$

Interpretation

The leading Hardy–Ramanujan approximation is the first approximation to $P_+(x)$ after replacing N by n and discarding $1 - 1/x$. In contrast, [equations \(3.6\) and \(3.7\)](#) records every Rademacher arc, both growing and reflected, with its exact algebraic amplitude. It is therefore a complete exponentially improved expansion rather than merely an all-orders Poincaré series.

4 Ordering of the sectors and a direct tail estimate

For a fixed k , the growing sector has absolute exponential scale $e^{x/k}$ and the reflected sector has scale $e^{-x/k}$. Relative to $P_+(x)$, their exponential gaps are

$$\delta_{k,+} = 1 - \frac{1}{k} \quad (k \geq 2), \quad \delta_{k,-} = 1 + \frac{1}{k} \quad (k \geq 2), \quad \delta_{1,-} = 2. \quad (4.1)$$

The first correction is therefore the $k = 2$ growing sector, of relative order $e^{-x/2}$, with parity amplitude $(-1)^n/\sqrt{2}$.

The following elementary estimate is less sharp than Lehmer’s classical remainder bounds but is sufficient to justify the hierarchy used in this article.

Theorem 4.1 (Fixed-level Rademacher tail). *Let $K \geq 1$ be fixed and let $\mathcal{R}_K(n)$ be the part of equation (3.2) with $k > K$. As $x \rightarrow \infty$,*

$$\mathcal{R}_K(n) = O_K\left(x^{-1/2}e^{x/(K+1)}\right), \quad (4.2)$$

and therefore

$$\frac{\mathcal{R}_K(n)}{P_+(x)} = O_K\left(x^{3/2}e^{-(1-1/(K+1))x}\right). \quad (4.3)$$

Proof. Use $|A_k(n)| \leq k$. For $K < k \leq x$, we have $k/x \leq 1$, hence the brace in equation (3.2) is bounded in absolute value by a constant multiple of $e^{x/k} \leq e^{x/(K+1)}$. The sum of the prefactors is

$$x^{-2} \sum_{k \leq x} \sqrt{k} = O(x^{-1/2}),$$

which proves the desired bound for this range.

For $k > x$, put $z = x/k \in (0, 1)$. The brace is

$$2 \left(\cosh z - \frac{\sinh z}{z} \right) = O(z^2)$$

uniformly for $0 < z \leq 1$. Its full k th contribution is therefore

$$O\left(x^{-2} \sqrt{k} \frac{x^2}{k^2}\right) = O(k^{-3/2}).$$

Summing over $k > x$ gives $O(x^{-1/2})$, which is absorbed by equation (4.2). Dividing by $P_+(x) \asymp x^{-2}e^x$ proves equation (4.3). $\square$

Remark 4.2. For a specified residue class, some amplitudes $A_k(n)$ may vanish, so the first nonzero omitted sector can be smaller than the generic estimate. Lehmer's bounds and subsequent refinements provide much sharper constants. The role of theorem 4.1 is structural: every fixed Rademacher level is exponentially accurate relative to the dominant sector.

5 All-order coefficients in the unshifted variable

The shifted variable x makes each sector elementary, but many applications prefer the conventional scale $n^{-1/2}$. The infinite algebraic coefficient arrays arise entirely from re-expanding the shift $N = n - 1/24$.

Put

$$t = n^{-1/2}, \quad a = \frac{1}{24}, \quad N = n(1 - at^2), \quad x = \frac{\lambda}{t} \sqrt{1 - at^2}. \quad (5.1)$$

For $\sigma \in \{+1, -1\}$, define

$$C_{k,\sigma}(t) = (1 - at^2)^{-1} \exp\left[\frac{\sigma\lambda}{kt} \left(\sqrt{1 - at^2} - 1\right)\right] \left(1 - \frac{\sigma kt}{\lambda\sqrt{1 - at^2}}\right). \quad (5.2)$$

The apparent singularity at $t = 0$ is removable because $\sqrt{1 - at^2} - 1 = O(t^2)$.

Theorem 5.1 (Universal sector coefficient generator). *For every fixed $k \geq 1$ and $\sigma \in \{+1, -1\}$, let*

$$C_{k,\sigma}(t) = \sum_{m \geq 0} c_{k,\sigma,m} t^m, \quad c_{k,\sigma,0} = 1. \quad (5.3)$$

Then the corresponding term in the exact Rademacher expansion is

$$P_{k,\sigma}(n) = \frac{A_k(n)}{4\sqrt{3k}n} e^{\sigma\lambda\sqrt{n}/k} \sum_{m \geq 0} c_{k,\sigma,m} n^{-m/2}. \quad (5.4)$$

The series in equation (5.3) is the Taylor series of equation (5.2); in particular it converges for all sufficiently small $|t|$ and supplies an asymptotic expansion to arbitrary order.

Proof. The (k, σ) term in equation (3.2) is

$$\frac{A_k(n)}{4\sqrt{3k}N} e^{\sigma x/k} \left(1 - \frac{\sigma k}{x}\right).$$

Insert equation (5.1), factor out $A_k(n)e^{\sigma\lambda/tk}/(4\sqrt{3k}n)$, and collect the remaining factors. The result is exactly equation (5.2). Analyticity at $t = 0$ follows from the removable singularity just noted and the binomial series for $\sqrt{1 - at^2}$. $\square$

5.1 Logarithmic coefficients and a Bell-polynomial formula

Write

$$\log C_{k,\sigma}(t) = \sum_{j \geq 1} \ell_{k,\sigma,j} t^j. \quad (5.5)$$

All $\ell_{k,\sigma,j}$ can be written as finite sums.

Theorem 5.2 (Explicit logarithmic coefficients). *For $j \geq 1$,*

$$\begin{aligned} \ell_{k,\sigma,j} = & \mathbf{1}_{2|j} \frac{a^{j/2}}{j/2} \\ & + \mathbf{1}_{j \text{ odd}} \frac{\sigma\lambda}{k} \binom{1/2}{(j+1)/2} (-a)^{(j+1)/2} \\ & - \sum_{\substack{1 \leq q \leq j \\ j-q \text{ even}}} \frac{1}{q} \left(\frac{\sigma k}{\lambda}\right)^q \frac{(q/2)(j-q)/2}{((j-q)/2)!} a^{(j-q)/2}. \end{aligned} \quad (5.6)$$

Consequently,

$$c_{k,\sigma,m} = \frac{1}{m!} Y_m(1! \ell_{k,\sigma,1}, 2! \ell_{k,\sigma,2}, \dots, m! \ell_{k,\sigma,m}). \quad (5.7)$$

Proof. Taking the logarithm of equation (5.2) gives

$$\begin{aligned} \log C_{k,\sigma}(t) = & -\log(1 - at^2) + \frac{\sigma\lambda}{kt} (\sqrt{1 - at^2} - 1) \\ & + \log\left(1 - \frac{\sigma kt}{\lambda} (1 - at^2)^{-1/2}\right). \end{aligned}$$

The first term contributes a^r/r at degree $2r$. The second contributes

$$\frac{\sigma\lambda}{k} \binom{1/2}{r} (-a)^r$$

at degree $2r - 1$. For the third term, use

$$\log(1 - u) = - \sum_{q \geq 1} \frac{u^q}{q},$$

$$(1 - at^2)^{-q/2} = \sum_{r \geq 0} \frac{(q/2)_r}{r!} a^r t^{2r}.$$

Setting $j = q + 2r$ gives [equation \(5.6\)](#). Formula [equation \(5.7\)](#) is the exponential formula [equation \(2.9\)](#). $\square$

Corollary 5.3 (Triangular coefficient recurrence). *The same coefficients satisfy*

$$c_{k,\sigma,0} = 1, \quad c_{k,\sigma,m} = \frac{1}{m} \sum_{j=1}^m j \ell_{k,\sigma,j} c_{k,\sigma,m-j} \quad (m \geq 1). \quad (5.8)$$

Proof. Differentiate $C_{k,\sigma} = \exp(\sum_{j \geq 1} \ell_{k,\sigma,j} t^j)$ and compare coefficients in $C' = (\log C)'C$. $\square$

Remark 5.4 (Coefficient-field structure). Every $c_{k,\sigma,m}$ lies in $\mathbb{Q}[\lambda, \lambda^{-1}, k, k^{-1}]$ after imposing $\sigma^2 = 1$ and $a = 1/24$. More precisely, the parity of m constrains the powers of λ : even m gives a polynomial in λ^2 and λ^{-2} , while odd m has an overall factor compatible with $\sigma \lambda^{2r+1}$ or $\sigma \lambda^{-(2r+1)}$. This follows directly from [equation \(5.6\)](#) and the Bell formula.

5.2 The dominant Poincare expansion

Only the $(k, \sigma) = (1, +1)$ sector survives at every algebraic order relative to the leading exponential. Put

$$\omega_m = c_{1,+,m}. \quad (5.9)$$

Then

$$p(n) \sim \frac{e^{\lambda \sqrt{n}}}{4\sqrt{3}n} \sum_{m \geq 0} \omega_m n^{-m/2}. \quad (5.10)$$

For each fixed M the difference after truncation at $m = M - 1$ is the sum of an algebraic remainder of order $n^{-M/2}$ in the dominant sector and an exponentially small Rademacher remainder. The all-order statement with explicit error estimates is studied in [\[13, 2\]](#).

The first coefficients are

$$\begin{aligned} \omega_0 &= 1, \\ \omega_1 &= -\frac{\sqrt{6}}{2\pi} - \frac{\sqrt{6}\pi}{144}, \\ \omega_2 &= \frac{1}{16} + \frac{\pi^2}{6912}, \\ \omega_3 &= -\frac{\sqrt{6}}{32\pi} - \frac{\sqrt{6}\pi}{2304} - \frac{\sqrt{6}\pi^3}{2985984}, \\ \omega_4 &= \frac{5}{1536} + \frac{5\pi^2}{497664} + \frac{\pi^4}{286654464}. \end{aligned} \quad (5.11)$$

The formulas above are useful for checking implementations, but the finite closed sum in the next section is better for direct evaluation.

6 A finite closed formula for the dominant coefficients

O’Sullivan’s coefficient formula can be derived directly from the generating function [equation \(5.2\)](#). The derivation is included because it reveals a compact Catalan substitution and links the partition expansion to Lagrange inversion.

Theorem 6.1 (Finite formula for ω_m). *For every $m \geq 0$,*

$$\omega_m = \frac{1}{(-4\sqrt{6})^m} \sum_{j=0}^{\lfloor (m+1)/2 \rfloor} \binom{m+1}{j} \frac{m+1-j}{(m+1-2j)!} \left(\frac{\pi}{6}\right)^{m-2j}. \quad (6.1)$$

Terms with a negative factorial index are understood to be absent. Formula [equation \(6.1\)](#) is equivalent to the coefficient expression in [\[13, 2\]](#).

Proof. Let

$$r = \frac{t}{4\sqrt{6}}, \quad A = \frac{\pi}{6}, \quad s = \sqrt{1-4r^2}, \quad w = \frac{1-s}{2r}. \quad (6.2)$$

Then

$$w = r(1+w^2), \quad r = \frac{w}{1+w^2}, \quad s = \frac{1-w^2}{1+w^2}. \quad (6.3)$$

A direct substitution into $C_{1,+}(t)$ gives

$$C_{1,+}(4\sqrt{6}r) = e^{-Aw} \frac{(1+w^2)^2}{(1-w^2)^2} \left(1 - \frac{2w}{A(1-w^2)}\right). \quad (6.4)$$

Since

$$\frac{dw}{dr} = \frac{(1+w^2)^2}{1-w^2},$$

the right-hand side of [equation \(6.4\)](#) is

$$-\frac{1}{A} \frac{d}{dr} \left(\frac{e^{-Aw}}{1-w^2} \right). \quad (6.5)$$

Coefficient extraction therefore yields

$$[r^m]C_{1,+}(4\sqrt{6}r) = -\frac{m+1}{A} [r^{m+1}] \frac{e^{-Aw}}{1-w^2}. \quad (6.6)$$

To extract the remaining coefficient, use $r = w/(1+w^2)$ in Cauchy's coefficient integral. The Jacobian is

$$dr = \frac{1-w^2}{(1+w^2)^2} dw,$$

so the factor $1-w^2$ cancels and

$$[r^{m+1}] \frac{e^{-Aw}}{1-w^2} = [w^{m+1}] e^{-Aw} (1+w^2)^m. \quad (6.7)$$

Expanding the last expression gives

$$\begin{aligned} [r^m]C_{1,+}(4\sqrt{6}r) &= -\frac{m+1}{A} \sum_{j=0}^{\lfloor (m+1)/2 \rfloor} \binom{m}{j} \frac{(-A)^{m+1-2j}}{(m+1-2j)!} \\ &= (-1)^m \sum_{j=0}^{\lfloor (m+1)/2 \rfloor} \binom{m+1}{j} \frac{m+1-j}{(m+1-2j)!} A^{m-2j}, \end{aligned}$$

where we used $(m+1)\binom{m}{j} = (m+1-j)\binom{m+1}{j}$. Finally, $[t^m] = (4\sqrt{6})^{-m}[r^m]$, proving [equation \(6.1\)](#). $\square$

Remark 6.2 (Why a finite sum exists). The square-root shift might suggest an infinite convolution at each order. The Catalan coordinate w absorbs that square root, and the Rademacher derivative creates the exact derivative in [equation \(6.5\)](#). After the coefficient-change Jacobian cancels, only the polynomial $(1+w^2)^m$ remains. This is the structural reason the coefficient at a fixed order is a finite binomial sum.

7 The multiplicative reciprocal $1/p(n)$

Although the main inverse in this article is functional, the reciprocal transseries is an immediate corollary of the exact factorization [equation \(3.6\)](#).

Theorem 7.1 (Exact reciprocal transseries). *For x sufficiently large that $|\mathcal{E}(x; n)| < 1$,*

$$\frac{1}{p(n)} = \frac{x^2}{C} e^{-x} \frac{1}{1-x^{-1}} \sum_{r \geq 0} (-1)^r \mathcal{E}(x; n)^r. \quad (7.1)$$

This identity is convergent. As a formal transseries, it remains valid without the analytic inequality.

Proof. Invert [equation \(3.6\)](#) and expand $(1 + \mathcal{E})^{-1}$ geometrically. $\square$

At a finite Rademacher level, write

$$\mathcal{E}_K(x; n) = \sum_{j \in J_K} g_j(x; n) q_j, \quad (7.2)$$

where the q_j are independent exponential variables. For a multi-index $\alpha = (\alpha_j)_{j \in J_K}$, put $|\alpha| = \sum_j \alpha_j$, $\alpha! = \prod_j \alpha_j!$, and $q^\alpha = \prod_j q_j^{\alpha_j}$.

Corollary 7.2 (Closed multinomial coefficient formula). *At finite level K ,*

$$[q^\alpha](1 + \mathcal{E}_K)^{-1} = (-1)^{|\alpha|} \frac{|\alpha|!}{\alpha!} \prod_{j \in J_K} g_j(x; n)^{\alpha_j}. \quad (7.3)$$

Proof. Only the term $(-1)^{|\alpha|} \mathcal{E}_K^{|\alpha|}$ in the geometric series can produce q^α , and the multinomial theorem gives the stated coefficient. $\square$

For the ordinary Poincaré expansion, define

$$\left(\sum_{m \geq 0} \omega_m t^m \right)^{-1} = \sum_{m \geq 0} \eta_m t^m. \quad (7.4)$$

Then

$$\eta_0 = 1, \quad \eta_m = - \sum_{j=1}^m \omega_j \eta_{m-j}, \quad (7.5)$$

and

$$\frac{1}{p(n)} \sim 4\sqrt{3} n e^{-\lambda\sqrt{n}} \sum_{m \geq 0} \eta_m n^{-m/2}. \quad (7.6)$$

Equivalently, if $\log(\sum \omega_m t^m) = \sum \ell_{1,+} t^j$, then

$$\eta_m = \frac{1}{m!} Y_m(-1! \ell_{1,+}, \dots, -m! \ell_{1,+}, m). \quad (7.7)$$

Thus reciprocal coefficients require no new analytic input.

8 What is the inverse of a discrete sequence?

The values $p(0) = p(1) = 1$ coincide, but $p(n)$ is strictly increasing for $n \geq 1$. Indeed, appending a part 1 injects the partitions of n into those of $n + 1$, and the one-part partition $(n + 1)$ is not in the image. Hence the generalized inverse

$$\nu(y) = \min\{n \geq 1 : p(n) \geq y\}, \quad y \geq 1, \quad (8.1)$$

is well defined and integer valued.

There are three related objects, which should not be conflated.

1. *The exact generalized inverse ν .* This is a step function. It has unit jumps and therefore requires an explicit rounding or sawtooth layer; no purely smooth transseries can represent those jumps uniformly to $o(1)$.
2. *The inverse along the range.* If $y = p(n)$, the desired answer is the integer n . A smooth asymptotic expression can approximate it with exponentially small error, after which nearest-integer rounding is exact for all sufficiently large n .
3. *The inverse of an interpolation.* Any strictly increasing continuous interpolation $\mathcal{P}$ with $\mathcal{P}(n) = p(n)$ has an ordinary inverse $\mathcal{P}^{-1}$. Then

$$\nu(y) = \lceil \mathcal{P}^{-1}(y) \rceil. \quad (8.2)$$

Different interpolations differ within the final unit-scale layer, so their smooth inverses are not canonically identical.

The Rademacher coefficients create a second issue: $A_k(n)$ is periodic rather than slowly varying. A naive replacement of n by a real variable inside the trigonometric formula gives one possible analytic interpolation, but not a unique one. Our solution is intrinsic at each exponential level: fix the residue of n modulo the periods of the retained sectors. The resulting finite-level function is analytic in the phase variable x and has a canonical asymptotic inverse. This construction is developed in [section 11](#).

Scope of the inverse formulas

The formulas through [section 10](#) are phase independent and therefore apply directly to large real ordinates y . The exponentially small formulas in [sections 11](#) and [12](#) describe the inverse along a specified arithmetic phase, in particular along $y = p(n)$ after the residue class of n is known. The phase can itself be recovered by rounding the phase-independent dominant inverse for all sufficiently large exact ordinates; see [theorem 13.3](#).

9 Exact Lambert carrier and its polynomial–logarithmic expansion

The first inversion step retains only

$$y = C \frac{e^X}{X^2}. \quad (9.1)$$

This differs from the complete dominant sector $P_+(x)$ by the factor $1 - 1/x$, which will be restored in [section 10](#).

Theorem 9.1 (Exact Lambert carrier). *For sufficiently large y , the large positive solution of [equation \(9.1\)](#) is*

$$X(y) = -2W_{-1}\left(-\frac{1}{2}\sqrt{\frac{C}{y}}\right). \quad (9.2)$$

Equivalently, with

$$R = \log \frac{y}{C}, \quad L = \log R, \quad (9.3)$$

we have

$$X - 2 \log X = R. \quad (9.4)$$

Proof. Equation (9.1) implies

$$\sqrt{\frac{C}{y}} = X e^{-X/2}.$$

Multiplying by $-1/2$ gives

$$-\frac{1}{2} \sqrt{\frac{C}{y}} = \left(-\frac{X}{2}\right) e^{-X/2}.$$

The solution with $X \rightarrow +\infty$ corresponds to the real branch W_{-1} , proving (9.2). Taking logarithms of (9.1) proves (9.4). $\square$

The Lambert formula is already an exact closed form for the carrier. To obtain a pure logarithmic transseries, write $X = R(1 + v)$. Equation (9.4) becomes

$$v = z(L + \log(1 + v)), \quad z = \frac{2}{R}. \quad (9.5)$$

This is in the standard Lagrange form $v = z\phi(v)$.

Theorem 9.2 (All polynomial–logarithmic carrier coefficients). *As $y \rightarrow \infty$,*

$$X \sim R + 2L + \sum_{j \geq 1} \frac{P_j(L)}{R^j}, \quad (9.6)$$

where

$$P_j(L) = \frac{2^{j+1}}{j+1} [u^j] (L + \log(1 + u))^{j+1} \quad (9.7)$$

$$= 2^{j+1} \sum_{q=1}^j \frac{s(j, q)}{(j+1-q)!} L^{j+1-q}. \quad (9.8)$$

For every fixed J ,

$$X = R + 2L + \sum_{j=1}^J \frac{P_j(L)}{R^j} + O\left(\frac{L^{J+1}}{R^{J+1}}\right). \quad (9.9)$$

Proof. Lagrange–Bürmann inversion applied to (9.5) gives

$$v = \sum_{m \geq 1} \frac{z^m}{m} [u^{m-1}] (L + \log(1 + u))^m. \quad (9.10)$$

Multiplying by R and putting $m = j+1$ yields (9.7). Expand the power by the binomial theorem and use (2.11). Only $1 \leq q \leq j$ contributes to $[u^j]$, and simplification gives (9.8). The remainder estimate follows from the analytic implicit-function theorem applied to (9.5), with $z = 2/R$ and $L = \log R$ treated on the asymptotic scale $L/R \rightarrow 0$. $\square$

The first polynomials are

$$\begin{aligned} P_1(L) &= 4L, \\ P_2(L) &= -4L(L-2), \end{aligned}$$

$$\begin{aligned}
 P_3(L) &= \frac{8}{3}L(2L^2 - 9L + 6), \\
 P_4(L) &= -\frac{8}{3}L(3L^3 - 22L^2 + 36L - 12), \\
 P_5(L) &= \frac{16}{15}L(12L^4 - 125L^3 + 350L^2 - 300L + 60).
 \end{aligned} \tag{9.11}$$

Thus, for example,

$$\begin{aligned}
 X &= R + 2L + \frac{4L}{R} + \frac{-4L^2 + 8L}{R^2} + \frac{\frac{16}{3}L^3 - 24L^2 + 16L}{R^3} \\
 &\quad + \frac{-8L^4 + \frac{176}{3}L^3 - 96L^2 + 32L}{R^4} + \cdots.
 \end{aligned} \tag{9.12}$$

Remark 9.3 (Two useful normalizations). One may instead expand $-W_{-1}(-\varepsilon)$ in $H = \log(1/\varepsilon)$ and $\log H$. Formula [equation \(9.8\)](#) is tailored to the partition problem because $R = \log(y/C)$ makes the phase equation exactly $X - 2 \log X = R$ and avoids additional constant translations.

10 Inverting the full dominant Rademacher sector

We now solve

$$y = P_+(x) = Ce^x x^{-2} \left(1 - \frac{1}{x}\right). \tag{10.1}$$

The Lambert carrier X solves the same equation without the final factor. The correction is algebraic in $1/X$.

Theorem 10.1 (Rational correction series). *Let $X = X(y)$ be given by [equation \(9.2\)](#), put $z = X^{-1}$, and let $U(z)$ be the unique formal series in $z\mathbb{Q}[[z]]$ satisfying*

$$\boxed{U + \log(1 - z + zU) - 3 \log(1 + zU) = 0.} \tag{10.2}$$

Then the asymptotic inverse of [equation \(10.1\)](#) is

$$x_0(y) = X + U(X^{-1}). \tag{10.3}$$

Moreover, U is analytic in a neighborhood of $z = 0$, so its series has a positive radius of convergence.

Proof. Put $x = X + U$ and $z = 1/X$. Dividing [equation \(10.1\)](#) by the carrier identity $y = Ce^X X^{-2}$ gives

$$e^U (1 + zU)^{-2} \left(1 - \frac{z}{1 + zU}\right) = 1.$$

Since

$$1 - \frac{z}{1 + zU} = \frac{1 - z + zU}{1 + zU},$$

taking logarithms gives [equation \(10.2\)](#). At $(z, U) = (0, 0)$ the left-hand side vanishes and its derivative with respect to U equals 1. The analytic implicit-function theorem therefore gives a unique analytic solution $U(z)$ near zero. Substitution recovers [equation \(10.1\)](#). $\square$

Theorem 10.2 (Triangular formula for the rational coefficients). *Write*

$$U(z) = \sum_{m \geq 1} b_m z^m, \quad U_{<m}(z) = \sum_{j=1}^{m-1} b_j z^j. \tag{10.4}$$

Then

$$b_m = -[z^m] \{ \log(1 - z + zU_{<m}) - 3 \log(1 + zU_{<m}) \}. \quad (10.5)$$

Equivalently,

$$\begin{aligned} b_m &= \sum_{q=1}^m \frac{(-1)^q}{q} [z^{m-q}] (U_{<m} - 1)^q \\ &\quad + 3 \sum_{q=1}^m \frac{(-1)^{q+1}}{q} [z^{m-q}] U_{<m}^q. \end{aligned} \quad (10.6)$$

In particular, every b_m is rational.

Proof. In [equation \(10.2\)](#), the coefficient of z^m contributed by U is b_m . Both logarithmic terms contain U only after multiplication by z , so their z^m coefficients involve only $b_1, \dots, b_{m-1}$. Solving the coefficient equation gives [equation \(10.5\)](#). Expanding the two logarithms gives [equation \(10.6\)](#). Rationality follows inductively. $\square$

The first fifteen coefficients are

$$\begin{aligned} b_1 &= 1, & b_2 &= \frac{5}{2}, & b_3 &= \frac{13}{3}, & b_4 &= \frac{53}{12}, \\ b_5 &= -\frac{14}{5}, & b_6 &= -\frac{763}{30}, & b_7 &= -\frac{2179}{35}, & b_8 &= -\frac{42289}{630}, \\ b_9 &= \frac{132973}{1260}, & b_{10} &= \frac{3449711}{5040}, & b_{11} &= \frac{29813143}{18480}, & b_{12} &= \frac{496569589}{356400}, \\ b_{13} &= -\frac{4334124833}{926640}, & b_{14} &= -\frac{509723063851}{21621600}, & b_{15} &= -\frac{298798016297}{5896800}. \end{aligned} \quad (10.7)$$

Thus

$$x_0 = X + \frac{1}{X} + \frac{5}{2X^2} + \frac{13}{3X^3} + \frac{53}{12X^4} - \frac{14}{5X^5} + \dots \quad (10.8)$$

10.1 The phase-independent inverse index

Define

$$n_+(y) = \frac{1}{24} + \frac{3x_0(y)^2}{2\pi^2}. \quad (10.9)$$

This is the inverse index supplied by the complete dominant positive Rademacher sector. It already includes every algebraic correction; the difference from an exact index n with $y = p(n)$ is beyond all algebraic orders.

Corollary 10.3 (All algebraic coefficients of the inverse index). *Let*

$$\beta_j = 3b_{j+1} + \frac{3}{2} \sum_{r=1}^{j-1} b_r b_{j-r} \quad (j \geq 1). \quad (10.10)$$

Then

$$n_+(y) = \frac{3X^2}{2\pi^2} + \left(\frac{1}{24} + \frac{3}{\pi^2} \right) + \frac{1}{\pi^2} \sum_{j \geq 1} \frac{\beta_j}{X^j}. \quad (10.11)$$

The first values are

$$\begin{aligned} \beta_1 &= \frac{15}{2}, & \beta_2 &= \frac{29}{2}, & \beta_3 &= \frac{83}{4}, & \beta_4 &= \frac{559}{40}, \\ \beta_5 &= -\frac{611}{20}, & \beta_6 &= -\frac{112459}{840}, & \beta_7 &= -\frac{101329}{420}, & \beta_8 &= -\frac{228677}{3360}. \end{aligned} \quad (10.12)$$

Proof. Insert $x_0 = X + \sum_{m \geq 1} b_m X^{-m}$ into [equation \(10.9\)](#). The term $2XU$ contributes $3b_{j+1}/(\pi^2 X^j)$, and U^2 contributes $3(\sum_{r=1}^{j-1} b_r b_{j-r})/(2\pi^2 X^j)$. $\square$

Remark 10.4 (The constant shift). The carrier-only index $1/24 + 3X^2/(2\pi^2)$ underestimates the exact index by a quantity tending to $3/\pi^2$. The first correction $U \sim X^{-1}$ contributes exactly $3/\pi^2$ to the inverse index. This explains the approximately -0.304 carrier error visible in the numerical table of [section 14](#).

11 The full arithmetic inverse transseries

The remaining correction to $n_+(y)$ is exponentially small but arithmetically oscillatory. The oscillation must be frozen before one can speak of an analytic inverse.

11.1 Finite Rademacher levels and residue classes

For $K \geq 1$, put

$$L_K = \text{lcm}(1, 2, \dots, K). \quad (11.1)$$

Fix $r \in \mathbb{Z}/L_K\mathbb{Z}$. Since $A_k(n)$ is periodic modulo k , the values $A_k(r)$ are unambiguously defined for all $k \leq K$. Define the finite-level analytic function

$$P_{K,r}(x) = \frac{C}{x^2} \sum_{k=1}^K \frac{A_k(r)}{\sqrt{k}} \left\{ e^{x/k} \left(1 - \frac{k}{x} \right) + e^{-x/k} \left(1 + \frac{k}{x} \right) \right\}. \quad (11.2)$$

For large x , the $k = 1$ positive term dominates, so $P'_{K,r}(x) > 0$ and $P_{K,r}$ has a unique inverse branch tending to infinity. Denote it by $x_{K,r}(y)$.

Relative to P_+ , write

$$P_{K,r}(x) = P_+(x) (1 + \mathcal{E}_{K,r}(x)), \quad (11.3)$$

where

$$\begin{aligned} \mathcal{E}_{K,r}(x) &= e^{-2x} \frac{1+x^{-1}}{1-x^{-1}} \\ &\quad + \sum_{k=2}^K \frac{A_k(r)}{\sqrt{k}} \left\{ e^{-(1-1/k)x} \frac{1-k/x}{1-1/x} + e^{-(1+1/k)x} \frac{1+k/x}{1-1/x} \right\}. \end{aligned} \quad (11.4)$$

Introduce the dominant logarithmic phase

$$\Phi(x) = x - 2 \log x + \log \left(1 - \frac{1}{x} \right), \quad x > 1. \quad (11.5)$$

Then $x_0 = h(R)$, where $R = \log(y/C)$ and $h = \Phi^{-1}$, while the full finite-level inverse solves

$$\Phi(x) + \Psi_{K,r}(x) = R, \quad \Psi_{K,r}(x) = \log(1 + \mathcal{E}_{K,r}(x)). \quad (11.6)$$

The slope divisor is

$$D(x) = \Phi'(x) = 1 - \frac{2}{x} + \frac{1}{x(x-1)} = \frac{x^2 - 3x + 3}{x(x-1)} > 0. \quad (11.7)$$

More generally,

$$\Phi^{(m)}(x) = \mathbf{1}_{m=1} + (-1)^{m-1} (m-1)! ((x-1)^{-m} - 3x^{-m}), \quad m \geq 1. \quad (11.8)$$

11.2 Independent exponential variables

Let

$$J_K = \{(1, -)\} \cup \{(k, +), (k, -) : 2 \leq k \leq K\}. \quad (11.9)$$

For $j \in J_K$, define the gap δ_j and amplitude g_j by

$$\delta_{1,-} = 2, \quad g_{1,-}(x) = \frac{1+x^{-1}}{1-x^{-1}}, \quad (11.10)$$

$$\delta_{k,+} = 1 - \frac{1}{k}, \quad g_{k,+}(x; r) = \frac{A_k(r)}{\sqrt{k}} \frac{1 - k/x}{1 - 1/x}, \quad (11.11)$$

$$\delta_{k,-} = 1 + \frac{1}{k}, \quad g_{k,-}(x; r) = \frac{A_k(r)}{\sqrt{k}} \frac{1 + k/x}{1 - 1/x}. \quad (11.12)$$

Introduce independent variables q_j and write

$$\mathcal{E}_{K,r}(x; \mathbf{q}) = \sum_{j \in J_K} q_j g_j(x; r). \quad (11.13)$$

At the end of the computation one specializes

$$q_j = e^{-\delta_j x_0}. \quad (11.14)$$

When $x = x_0 + v$, the actual exponential is $q_j e^{-\delta_j v}$, so define

$$\Psi_{K,r}(x_0 + v; \mathbf{q}) = \log \left(1 + \sum_{j \in J_K} q_j g_j(x_0 + v; r) e^{-\delta_j v} \right). \quad (11.15)$$

This formalization treats resonant exponent sums correctly: distinct products are first kept as distinct multivariate monomials and may be combined only after [equation \(11.14\)](#).

11.3 The triangular Newton–Hensel recurrence

Choose any monomial order on $\mathbb{N}^{J_K}$ that is compatible with addition, for example graded lexicographic order. For a multi-index α , write $\mathbf{q}^\alpha = \prod_j q_j^{\alpha_j}$. Seek

$$v(\mathbf{q}) = \sum_{\alpha \neq 0} V_\alpha(x_0; r) \mathbf{q}^\alpha. \quad (11.16)$$

For a fixed α , let $v_{<\alpha}$ be the sum of terms preceding α in the chosen order.

Theorem 11.1 (Complete multivariate inverse-coefficient recurrence). *Let*

$$\kappa_m(x_0) = \frac{\Phi^{(m)}(x_0)}{m!}, \quad \kappa_1(x_0) = D(x_0). \quad (11.17)$$

The unique formal solution of [equation \(11.6\)](#) with $x = x_0 + v$ is given recursively by

$$V_\alpha = -\frac{1}{D(x_0)} [\mathbf{q}^\alpha] \left\{ \sum_{m \geq 2} \kappa_m(x_0) v_{<\alpha}^m + \Psi_{K,r}(x_0 + v_{<\alpha}; \mathbf{q}) \right\}. \quad (11.18)$$

At each α , the right-hand side is a finite expression involving only previously computed coefficients. It is a rational combination of x_0 , $(x_0 - 1)^{-1}$, the gaps δ_j , and the arithmetic amplitudes $A_k(r)/\sqrt{k}$.

Proof. Taylor expansion of [equation \(11.6\)](#) at x_0 gives

$$D(x_0)v + \sum_{m \geq 2} \kappa_m(x_0)v^m + \Psi_{K,r}(x_0 + v; \mathbf{q}) = 0. \quad (11.19)$$

The only term in which V_α can occur linearly at monomial $\mathbf{q}^\alpha$ is $D(x_0)v$. Every term v^m with $m \geq 2$ uses a sum of at least two positive multi-indices and therefore depends only on lower coefficients. Every term in Ψ contains at least one explicit q_j ; if V_α appears inside v , multiplication by that q_j raises the multi-index beyond α . Hence the coefficient equation at α is triangular and solving it gives [equation \(11.18\)](#). Finiteness follows because there are only finitely many variables and a fixed multi-index has finitely many additive decompositions. $\square$

Corollary 11.2 (First coefficient in each exponential direction). *Let e_j be the unit multi-index of the variable q_j . Then*

$$V_{e_j}(x_0; r) = -\frac{g_j(x_0; r)}{D(x_0)}. \quad (11.20)$$

After [equation \(11.14\)](#), the first-order contribution of sector j is

$$v_j^{(1)}(y; r) = -\frac{g_j(x_0; r)}{D(x_0)} e^{-\delta_j x_0}. \quad (11.21)$$

11.4 A closed Lagrange–Bürmann formula

The triangular recurrence is ideal for exact coefficient production. A second formula packages all orders into one operator identity.

Theorem 11.3 (Lagrange–Bürmann formula for the full inverse). *Let $h = \Phi^{-1}$, $R = \log(y/C)$, and*

$$A(s) = \Psi_{K,r}(h(s)). \quad (11.22)$$

Then, as a formal transseries in the exponential variables,

$$S = (I + A)^{-1}(R) = R + \sum_{m \geq 1} \frac{(-1)^m}{m!} \left(\frac{d}{dR} \right)^{m-1} A(R)^m, \quad (11.23)$$

and

$$\boxed{x_{K,r}(y) = h(S)}. \quad (11.24)$$

Equations [equations \(11.23\)](#) and [\(11.24\)](#) are an all-orders closed formula for the inverse transseries.

Proof. Put $s = \Phi(x)$. Then $x = h(s)$ and [equation \(11.6\)](#) becomes

$$s + A(s) = R.$$

The classical Lagrange–Bürmann reversion formula for the near-identity map $s \mapsto s + A(s)$ gives [equation \(11.23\)](#). Composing with h gives [equation \(11.24\)](#). $\square$

To turn [equation \(11.24\)](#) into coefficient polynomials, introduce a bookkeeping parameter τ by replacing A with τA , and set

$$d_m(R) = \frac{(-1)^m}{m!} \left(\frac{d}{dR} \right)^{m-1} A(R)^m. \quad (11.25)$$

Then $S - R = \sum_{m \geq 1} d_m \tau^m$. Faà di Bruno’s formula gives the following explicit Bell-polynomial coefficient.

Corollary 11.4 (Coefficient at perturbative order m). *If*

$$x_{K,r}(y; \tau) = x_0 + \sum_{m \geq 1} v_m \tau^m, \quad (11.26)$$

then

$$v_m = \frac{1}{m!} \sum_{q=1}^m h^{(q)}(R) B_{m,q}(1!d_1, 2!d_2, \dots, (m-q+1)!d_{m-q+1}). \quad (11.27)$$

The derivatives of h are generated recursively by

$$h'(R) = \frac{1}{\Phi'(x_0)}, \quad (11.28)$$

$$h^{(m)}(R) = -\frac{1}{\Phi'(x_0)} \sum_{q=2}^m \Phi^{(q)}(x_0) B_{m,q}(h', h'', \dots, h^{(m-q+1)}), \quad m \geq 2. \quad (11.29)$$

Proof. Expand $h(R + \sum d_m \tau^m)$ by Taylor's theorem and use the defining generating identity [equation \(2.7\)](#) for the coefficient of a power of $\sum d_m \tau^m$. For [equation \(11.29\)](#), differentiate $\Phi(h(R)) = R$ m times and solve the Faà di Bruno identity for the term containing $h^{(m)}$. $\square$

11.5 Low perturbative orders in differential form

For checking symbolic implementations, consider the one-parameter equation

$$\Phi(x) + \tau \Psi(x) = R, \quad x = x_0 + \tau v_1 + \tau^2 v_2 + \tau^3 v_3 + \dots \quad (11.30)$$

All functions and derivatives below are evaluated at x_0 , and $D = \Phi'(x_0)$. Direct coefficient comparison gives

$$v_1 = -\frac{\Psi}{D}, \quad (11.31)$$

$$v_2 = \frac{\Psi \Psi'}{D^2} - \frac{\Phi'' \Psi^2}{2D^3}, \quad (11.32)$$

$$v_3 = -\frac{\Psi(\Psi')^2}{D^3} - \frac{\Psi^2 \Psi''}{2D^3} + \frac{3\Phi'' \Psi^2 \Psi'}{2D^4} + \frac{\Phi''' \Psi^3}{6D^4} - \frac{(\Phi'')^2 \Psi^3}{2D^5}. \quad (11.33)$$

These formulas collect many exponential monomials at once; [equation \(11.18\)](#) is preferable when individual sectors or resonances are required.

11.6 The profinite arithmetic phase

The residue systems are compatible because $L_K \mid L_{K+1}$. A compatible sequence

$$r_K \in \mathbb{Z}/L_K \mathbb{Z}, \quad r_{K+1} \equiv r_K \pmod{L_K}, \quad (11.34)$$

defines a profinite integer $\widehat{r} \in \widehat{\mathbb{Z}}$. Conversely, every $\widehat{r}$ supplies all amplitudes $A_k(\widehat{r})$ by reduction modulo k .

Theorem 11.5 (Arithmetic-phase organization). *The family of finite-level inverse transseries*

$$\{x_{K,r_K}(y)\}_{K \geq 1}$$

is compatible under truncation whenever the residues satisfy [equation \(11.34\)](#). Hence the complete arithmetic inverse is naturally a projective family indexed by $\widehat{r} \in \widehat{\mathbb{Z}}$. For an actual partition index n , the diagonal embedding $n \mapsto \widehat{n}$ gives its phase.

Proof. Truncating the $(K + 1)$ st construction removes all variables associated with $k = K + 1$. For every retained $k \leq K$, compatibility of the residues gives $A_k(r_{K+1}) = A_k(r_K)$. Thus the truncated defining equations, and consequently their unique triangular solutions, agree. $\square$

Remark 11.6 (Why the profinite parameter is not optional). At algebraic order, all indices share the same expansion. At relative order $e^{-x/2}$, parity appears; at order $e^{-2x/3}$, the residue modulo 3 appears; and so on. No finite real-valued smooth parameter records all these congruence classes, while a profinite integer does so canonically.

12 First arithmetic inverse sectors and the master formula

At finite level K , the first total degree in the exponential variables is especially transparent. Combining equations (11.10) to (11.12) and (11.21) gives

$$x_{K,r}(y) = x_0 - \frac{1}{D(x_0)} \left[e^{-2x_0} \frac{1 + x_0^{-1}}{1 - x_0^{-1}} + \sum_{k=2}^K \frac{A_k(r)}{\sqrt{k}} \left\{ e^{-(1-1/k)x_0} \frac{1 - k/x_0}{1 - 1/x_0} + e^{-(1+1/k)x_0} \frac{1 + k/x_0}{1 - 1/x_0} \right\} \right] + O_q(2), \quad (12.1)$$

where $O_q(2)$ denotes monomials of total degree at least two in the independent q_j .

The first two growing arithmetic sectors are

$$v_{2,+}^{(1)} = -\frac{(-1)^r}{\sqrt{2} D(x_0)} \frac{1 - 2/x_0}{1 - 1/x_0} e^{-x_0/2}, \quad (12.2)$$

$$v_{3,+}^{(1)} = -\frac{2 \cos(2\pi r/3 - \pi/18)}{\sqrt{3} D(x_0)} \frac{1 - 3/x_0}{1 - 1/x_0} e^{-2x_0/3}. \quad (12.3)$$

Thus parity enters at scale $e^{-x_0/2}$ and the residue modulo 3 at scale $e^{-2x_0/3}$. More generally, the residue modulo k first appears in the growing k th Rademacher sector at scale $e^{-(1-1/k)x_0}$.

Remark 12.1 (Accumulation near relative exponent one). There are infinitely many first-order growing sectors with gaps $1 - 1/k < 1$. Therefore an instruction such as “retain every term through e^{-x} ” does not select a finite set of Rademacher sectors. A meaningful truncation must specify a Rademacher level K , an arithmetic cutoff, or a combined analytic error criterion. This is another reason to keep the exact convergent sector sum and finite-level formal calculus side by side.

12.1 From phase to index

Let

$$n_{K,r}(y) = \frac{1}{24} + \frac{3x_{K,r}(y)^2}{2\pi^2}. \quad (12.4)$$

If $v = \sum_{\alpha \neq 0} V_\alpha q^\alpha$, then

$$n_{K,r} = n_+(y) + \frac{3x_0}{\pi^2} v + \frac{3}{2\pi^2} v^2. \quad (12.5)$$

Hence the coefficient of each exponential monomial is explicit.

Corollary 12.2 (Inverse-index coefficient formula). *For $\alpha \neq 0$, let N_α be the coefficient of q^α in $n_{K,r} - n_+$. Then*

$$N_\alpha = \frac{3}{\pi^2} \left(x_0 V_\alpha + \frac{1}{2} \sum_{\substack{\beta+\gamma=\alpha \\ \beta, \gamma \neq 0}} V_\beta V_\gamma \right). \quad (12.6)$$

In particular,

$$N_{e_j} = -\frac{3x_0}{\pi^2 D(x_0)} g_j(x_0; r). \quad (12.7)$$

Proof. Expand the square in [equation \(12.5\)](#) and compare coefficients. $\square$

Combining all layers produces the main inverse formula.

Theorem 12.3 (Master nested inverse transseries). *Fix a finite level K and a residue $r \pmod{L_K}$. Let*

$$\begin{aligned} X(y) &= -2W_{-1}\left(-\frac{1}{2}\sqrt{\frac{C}{y}}\right), \\ x_0(y) &= X(y) + U(X(y)^{-1}), \\ v_{K,r}(y) &= \sum_{\alpha \neq 0} V_\alpha(x_0; r) \exp\left(-x_0 \sum_{j \in J_K} \alpha_j \delta_j\right), \end{aligned}$$

where U is determined by [equation \(10.5\)](#) and the V_α by [equation \(11.18\)](#). Then

$$n_{K,r}(y) = \frac{1}{24} + \frac{3}{2\pi^2} (x_0(y) + v_{K,r}(y))^2. \quad (12.8)$$

Substituting the Stirling-number expansion [equations \(9.6\)](#) and [\(9.8\)](#) for $X(y)$ turns [equation \(12.8\)](#) into a transseries expressed solely in

$$R^{-1}, \quad \log R, \quad \exp[-\delta(R + 2 \log R + \cdots)], \quad R = \log(y/C),$$

with every coefficient generated by finite Bell-polynomial and Cauchy-product operations.

Proof. The first line inverts the elementary carrier, the second restores the full dominant positive sector, and the third is the unique finite-level exponential correction of [theorem 11.1](#). [Equation \(1.11\)](#) then gives the index. Closure under substitution follows from the coefficient formulas already proved. $\square$

13 Validity, truncation error, and discrete recovery

13.1 Finite-level inverse error along exact partition values

Let n be an integer, let $x = \lambda\sqrt{n-1/24}$, and fix the compatible residue $r \equiv n \pmod{L_K}$. The exact partition value satisfies

$$p(n) = P_{K,r}(x) + \mathcal{R}_K(n). \quad (13.1)$$

For large x , logarithmic derivatives of $P_{K,r}$ are bounded away from zero because the dominant term has slope tending to 1.

Theorem 13.1 (Error after inversion of K Rademacher levels). *For fixed K and $y = p(n)$,*

$$x_{K,r}(y) - x = O_K\left(x^{3/2} e^{-(1-1/(K+1))x}\right), \quad (13.2)$$

and

$$n_{K,r}(y) - n = O_K\left(x^{5/2} e^{-(1-1/(K+1))x}\right). \quad (13.3)$$

Proof. By [theorem 4.1](#), the omitted tail divided by the dominant sector is $O_K(x^{3/2}e^{-(1-1/(K+1))x})$. Taking logarithms changes this by only a comparable amount. The derivative of $\log P_{K,r}(x)$ tends to 1, so the mean-value theorem converts the logarithmic residual into the phase error [equation \(13.2\)](#). Finally, $n = 1/24 + 3x^2/(2\pi^2)$, whose derivative with respect to x is $3x/\pi^2$, proving [equation \(13.3\)](#). $\square$

The case $K = 1$ is particularly useful because it is phase independent.

Corollary 13.2 (Beyond-all-orders accuracy of the dominant inverse). *For $y = p(n)$,*

$$n_+(y) - n = O\left(x^{5/2}e^{-x/2}\right) = O\left(n^{5/4}e^{-\lambda\sqrt{n}/2}\right). \quad (13.4)$$

In particular, $n_+(p(n)) - n \rightarrow 0$ faster than every power of n^{-1} .

Theorem 13.3 (Eventual exact recovery by rounding). *There exists n_0 such that for every integer $n \geq n_0$,*

$$n = \text{nint}(n_+(p(n))), \quad (13.5)$$

where nint denotes nearest-integer rounding. Consequently, for sufficiently large exact ordinates $y = p(n)$, the arithmetic phase $\widehat{n}$ can be recovered before any exponentially small sector is evaluated.

Proof. By [equation \(13.4\)](#), the absolute error tends to zero. It is therefore smaller than $1/2$ for all sufficiently large n , at which point nearest-integer rounding is exact. $\square$

Remark 13.4. The theorem is asymptotic and does not claim an optimal threshold. Direct computation in [section 14](#) shows that the dominant-sector inverse is already extraordinarily accurate at modest n .

13.2 Residual certificates

An approximate inverse should be checked by substitution rather than by counting formal terms. Let

$$F_{K,r}(x) = \log P_{K,r}(x), \quad \rho(\tilde{x}) = F_{K,r}(\tilde{x}) - \log y. \quad (13.6)$$

Proposition 13.5 (A posteriori error certificate). *Suppose $F'_{K,r}(x) \geq m > 0$ on an interval containing the exact root $x_{K,r}(y)$ and an approximation $\tilde{x}$. Then*

$$|\tilde{x} - x_{K,r}(y)| \leq \frac{|\rho(\tilde{x})|}{m}. \quad (13.7)$$

The corresponding index error satisfies

$$|\tilde{n} - n_{K,r}(y)| \leq \frac{3}{2\pi^2} |\tilde{x} - x_{K,r}(y)| (|\tilde{x}| + |x_{K,r}(y)|). \quad (13.8)$$

Proof. The mean-value theorem gives $|F_{K,r}(\tilde{x}) - F_{K,r}(x_{K,r})| = F'_{K,r}(\xi)|\tilde{x} - x_{K,r}|$ for an intermediate ξ ; use the lower bound. The index estimate is the factorization of the difference of two squares. $\square$

This certificate applies equally to a truncated symbolic transseries and to a numerically refined root. It is particularly effective because $F'_{K,r}(x) \rightarrow 1$.

14 Numerical checks

The following computations use exact integer values of $p(n)$, exact rational Dedekind sums inside $A_k(n)$, and high-precision evaluation of the elementary sector formula. They are intended as verification of normalization and signs, not as substitutes for the proofs.

14.1 Forward truncations

Let $P_K(x; n)$ denote [equation \(11.2\)](#) evaluated with the actual amplitudes $A_k(n)$ and the exact phase $x = \lambda\sqrt{n - 1/24}$. The table shows the signed relative error $(P_K - p(n))/p(n)$.

n	$p(n)$	$K = 1$	$K = 2$	$K = 3$	$K = 5$
25	1,958	$1.0717 \cdot 10^{-3}$	$3.0045 \cdot 10^{-6}$	$-6.1015 \cdot 10^{-5}$	$-3.4866 \cdot 10^{-6}$
50	204,226	$-7.3074 \cdot 10^{-5}$	$3.9028 \cdot 10^{-6}$	$2.1050 \cdot 10^{-7}$	$-1.1350 \cdot 10^{-7}$
100	190,569,292	$-1.8220 \cdot 10^{-6}$	$8.6838 \cdot 10^{-9}$	$-4.9491 \cdot 10^{-9}$	$3.1454 \cdot 10^{-10}$
250	230,793,554,364,681	$-1.0760 \cdot 10^{-9}$	$7.2267 \cdot 10^{-13}$	$4.3021 \cdot 10^{-14}$	$2.1256 \cdot 10^{-15}$
500	2,300,165,032,574,323,995,027	$-2.4389 \cdot 10^{-13}$	$1.7537 \cdot 10^{-17}$	$-2.0017 \cdot 10^{-19}$	$-3.5933 \cdot 10^{-22}$

Table 1: Relative error of finite Rademacher sector sums. Because amplitudes can change sign, accuracy need not improve monotonically at every intermediate K , although the convergent sum tends to $p(n)$.

14.2 Inverse truncations

For $y = p(n)$, let $n_X = 1/24 + 3X^2/(2\pi^2)$ be the carrier inverse, let n_+ be the complete dominant-sector inverse, and let n_K be obtained by numerically inverting $P_K(x; n)$ with the true residue data. The table reports $n_{\text{approx}} - n$.

n	carrier n_X	dominant n_+	$K = 2$	$K = 3$	$K = 5$
50	$-3.5044 \cdot 10^{-1}$	$4.5102 \cdot 10^{-4}$	$-2.4089 \cdot 10^{-5}$	$-1.2992 \cdot 10^{-6}$	$7.0057 \cdot 10^{-7}$
100	$-3.3600 \cdot 10^{-1}$	$1.5378 \cdot 10^{-5}$	$-7.3293 \cdot 10^{-8}$	$4.1772 \cdot 10^{-8}$	$-2.6548 \cdot 10^{-9}$
500	$-3.1768 \cdot 10^{-1}$	$4.4042 \cdot 10^{-12}$	$-3.1668 \cdot 10^{-16}$	$3.6146 \cdot 10^{-18}$	$6.4888 \cdot 10^{-21}$
1000	$-3.1356 \cdot 10^{-1}$	$4.2960 \cdot 10^{-17}$	$-3.1780 \cdot 10^{-23}$	$8.7815 \cdot 10^{-26}$	$-9.9717 \cdot 10^{-29}$

Table 2: Errors in inverse indices at exact ordinates $y = p(n)$. Restoring the factor $1 - 1/x$ removes the carrier's order-one bias; the remaining errors follow the exponentially small Rademacher hierarchy.

The parity-sensitive $k = 2$ correction explains why the sign of the first exponentially small error alternates. The very small dominant-sector errors at $n = 500$ and 1000 also illustrate [theorem 13.3](#): exact integer recovery requires no knowledge of the residue phase once the error is below one half.

15 Constructive coefficient algorithms

The formulas above are designed to be executable in a computer algebra system, but they require only formal power-series arithmetic. This section records deterministic algorithms and their dependency structure.

15.1 Forward coefficient array

Algorithm 15.1 (Coefficients of a Rademacher sector). *Given (k, σ, M) :*

1. For $1 \leq j \leq M$, evaluate the finite sum [equation \(5.6\)](#) to obtain $\ell_{k, \sigma, j}$.
2. Set $c_{k, \sigma, 0} = 1$.
3. For $1 \leq m \leq M$, use [equation \(5.8\)](#):

$$c_{k, \sigma, m} = \frac{1}{m} \sum_{j=1}^m j \ell_{k, \sigma, j} c_{k, \sigma, m-j}.$$

4. Return the sector [equation \(5.4\)](#) truncated after $m = M$.

The arithmetic complexity is $O(M^2)$ coefficient operations after the ℓ_j are available. If only the dominant sector is needed, the finite formula [equation \(6.1\)](#) evaluates each ω_m independently.

15.2 Lambert carrier and algebraic inverse

Algorithm 15.2 (Phase-independent inverse). Given a large ordinate y and algebraic order M :

1. Compute X from [equation \(9.2\)](#); alternatively, compute R, L and use [equation \(9.6\)](#) with P_j from [equation \(9.8\)](#).
2. Generate $b_1, \dots, b_M$ by [equation \(10.5\)](#).
3. Form $x_0 = X + \sum_{m=1}^M b_m X^{-m}$.
4. Return $n_+ = 1/24 + 3x_0^2/(2\pi^2)$.
5. Optionally, evaluate the residual

$$\rho = x_0 - 2 \log x_0 + \log(1 - x_0^{-1}) - \log(y/C)$$

and apply [theorem 13.5](#).

For an exact ordinate $y = p(n)$, nearest-integer rounding of the result is eventually exact.

15.3 Arithmetic exponential correction

Algorithm 15.3 (Finite-level full inverse). Given y , level K , compatible residue $r \pmod{L_K}$, and a finite set of desired multi-indices:

1. Compute x_0 by [theorem 15.2](#) or by numerical solution of [equation \(10.1\)](#).
2. Compute the exact rational Dedekind sums $s(h, k)$ and amplitudes $A_k(r)$ for $2 \leq k \leq K$.
3. Build the gaps and rational amplitudes in [equations \(11.10\) to \(11.12\)](#).
4. Traverse the multi-indices in a monomial order and compute V_α by [equation \(11.18\)](#).
5. Substitute $q_j = e^{-\delta_j x_0}$, form $x = x_0 + v$, and return $n = 1/24 + 3x^2/(2\pi^2)$.
6. Certify the result by the logarithmic residual of $P_{K,r}(x) - y$ and include the analytic Rademacher tail estimate if comparison with the exact sequence is required.

Remark 15.4 (Avoiding the apparent circularity). For an unknown exact ordinate $y = p(n)$, the residue r is initially unknown. Compute $n_+(y)$ first and round it. By [theorem 13.3](#), this recovers n for all sufficiently large inputs, after which every desired residue and Rademacher amplitude is known. Thus the arithmetic transseries refines an already identifiable integer; it is not needed to guess the phase blindly.

15.4 Coefficient reuse

The calculations have a useful modular architecture.

- The polynomials $P_j(L)$ depend only on the carrier exponent $X^{-2}e^X$.
- The rational numbers b_m depend only on the amplitude $1 - 1/x$.
- The inverse derivative polynomials $h^{(m)}$ depend only on the dominant phase Φ .
- The arithmetic data enter only through $A_k(r)$ in the sector amplitudes g_j .

Consequently, the universal coefficient tables may be precomputed once, while residue-dependent evaluations are inexpensive.

16 A general template for Rademacher-type sequences

The derivation is not limited to ordinary partitions. Suppose a sequence has an exact or exponentially accurate representation

$$a(n) = M(x(n)) \left(1 + \sum_{j \in J} c_j(n) g_j(x(n)) e^{-\delta_j x(n)} \right), \quad (16.1)$$

where

1. $x(n) \rightarrow \infty$ is invertible at the algebraic level;
2. $M(x) = C e^{\alpha x} x^{-\beta} A(x^{-1})$ with $A(0) = 1$;
3. $c_j(n)$ is periodic or otherwise belongs to a finite-dimensional phase system at each truncation level;
4. finite subsets of sectors have exponentially smaller tails.

Then the method of this article applies verbatim:

1. invert $C e^{\alpha x} x^{-\beta}$ with a Lambert W core;
2. invert the amplitude $A(x^{-1})$ by an ordinary implicit power series;
3. freeze the finite phase data and invert the remaining sectors by [equation \(11.18\)](#) or [equation \(11.24\)](#);
4. convert x back to the original index.

For the general carrier

$$y = C e^{\alpha X} X^{-\beta}, \quad \alpha, \beta > 0, \quad (16.2)$$

one obtains

$$X = -\frac{\beta}{\alpha} W_{-1} \left(-\frac{\alpha}{\beta} \left(\frac{C}{y} \right)^{1/\beta} \right). \quad (16.3)$$

The partition case is $(\alpha, \beta) = (1, 2)$. The same Bell-polynomial and triangular coefficient calculus therefore applies to many eta-quotient coefficients and Rademacher-type formulas, provided the relevant phase and tail hypotheses are verified.

Proposition 16.1 (General first exponential inverse correction). *Let $M_0(x)$ be a strictly increasing dominant approximation and suppose*

$$a = M_0(x) (1 + g(x) e^{-\delta x} + O(e^{-2\delta x})).$$

If x_0 solves $M_0(x_0) = a$, then

$$x = x_0 - \frac{g(x_0)}{(\log M_0)'(x_0)} e^{-\delta x_0} + O(e^{-2\delta x_0}) \quad (16.4)$$

up to the algebraic factors generated by differentiation.

Proof. Put $x = x_0 + v$, take logarithms, and retain the first order in both v and $e^{-\delta x_0}$. □

This proposition is the one-sector form of [equation \(11.21\)](#). The full multivariate recurrence is needed whenever products, resonances, or several periodic phases interact.

17 Structural consequences and research directions

17.1 Convergent perturbative blocks inside an exponentially complete object

The partition transseries has an uncommon hybrid structure. The exact Rademacher sum converges. Within every fixed sector, the unshifted coefficient series [equation \(5.2\)](#) is the Taylor series of an analytic

germ and therefore converges near $t = 0$. Likewise, the dominant inverse correction $U(z)$ is analytic near $z = 0$. Divergence is not needed to create the nonperturbative sectors: those sectors are already supplied by modularity and the circle method.

This illustrates a useful distinction:

- a *Poincaré expansion* records a single dominant sector and forgets all exponentially small information;
- a *transseries completion* records the other exponential sectors, whether or not the perturbative series in each sector diverges;
- an *exact transseries representation* may itself be convergent, as Rademacher’s formula is.

17.2 Arithmetic resurgence without analytic continuation in the index

The amplitudes $A_k(n)$ are arithmetic rather than Stokes constants. Nevertheless, they play a formally analogous role: they weight distinct exponential sectors and control exponentially small corrections to inversion. Their dependence on n is not naturally encoded by a single real analytic function. The profinite parameter of [theorem 11.5](#) provides a clean replacement: every finite exponential resolution sees only finitely much congruence information.

A promising next step is to formulate an “arithmetic transseries sheaf” whose base is $\widehat{\mathbb{Z}}$ and whose fibers are finite-level analytic transseries algebras. The projective compatibility proved here is the elementary case of such a structure.

17.3 Sharpening inverse error bounds

The crude estimate [equation \(13.3\)](#) uses only $|A_k(n)| \leq k$. Stronger bounds for Rademacher/Kloosterman sums and Lehmer’s explicit remainders should yield:

1. effective constants in the inverse error;
2. a certified minimal threshold for nearest-integer recovery from $n_+(p(n))$;
3. an adaptive rule choosing K from a target number of digits;
4. rigorous interval enclosures for the generalized inverse $v(y)$.

The residual method in [theorem 13.5](#) separates the local inversion error from the Rademacher tail, making such a refinement modular.

17.4 Canonical real interpolation

The finite-level residue interpolation is canonical along arithmetic subsequences, but it does not produce a unique global real interpolation of $p(n)$. Possible global choices include:

- trigonometric interpolation of each periodic A_k ;
- logarithmic piecewise-linear interpolation through the points $(n, \log p(n))$;
- a Newton series or cardinal interpolation satisfying prescribed growth conditions;
- analytic continuations based on Poincaré series or modular integrals.

Different choices may have the same algebraic inverse and the same values on the integer range while differing in exponentially small off-lattice terms. Classifying these differences is a natural nonuniqueness problem.

17.5 Formal verification targets

The exact Rademacher input now has an Isabelle/HOL formalization [6]. The remaining steps are well suited to staged formalization:

1. differentiation and algebraic normalization to [equation \(3.2\)](#);
2. the coefficient generator [equation \(5.2\)](#) and recurrence [equation \(5.8\)](#);
3. the Catalan/Lagrange proof of [equation \(6.1\)](#);
4. formal implicit-series construction of U and the recurrence [equation \(10.5\)](#);
5. finite multivariate triangular inversion [equation \(11.18\)](#);
6. residual-to-root error transfer under an explicit derivative lower bound.

The Bell-polynomial infrastructure described in the ProveIt combinatorial coefficient calculus is directly relevant to steps 2, 4, and 5.

18 Conclusion

The ordinary partition numbers admit a particularly explicit transseries description once Rademacher's exact formula is written in the shifted phase $x = \pi\sqrt{2(n-1/24)}/3$. The main conclusions are:

1. The full forward transseries is the exact convergent sector sum [equation \(3.2\)](#). Every sector has a two-term amplitude in x^{-1} .
2. Re-expansion in $n^{-1/2}$ is generated by the universal analytic function [equation \(5.2\)](#). Every coefficient is given by the Bell formula [equation \(5.7\)](#) or recurrence [equation \(5.8\)](#); the dominant coefficients also have the finite formula [equation \(6.1\)](#).
3. The elementary inverse carrier is exactly a W_{-1} expression. Its full logarithmic expansion has the Stirling-number coefficient polynomials [equation \(9.8\)](#).
4. Restoring the dominant amplitude produces a convergent rational correction $U(1/X)$ with the all-order recurrence [equation \(10.5\)](#).
5. The remaining inverse sectors are arithmetic. At finite Rademacher level and fixed residue class, every coefficient is generated by the triangular recurrence [equation \(11.18\)](#), equivalently by the Lagrange–Bürmann/Bell formula [equations \(11.24\)](#) and [\(11.27\)](#). Compatible levels are indexed by a profinite phase.
6. For exact ordinates $y = p(n)$, the phase-independent dominant inverse differs from n by an exponentially small quantity. It therefore recovers the exact integer by rounding for all sufficiently large n .

The resulting nested formula [equation \(12.8\)](#) is a complete constructive answer to the asymptotic inversion problem: it separates the universal Lambert/logarithmic geometry, the rational amplitude correction, and the periodic arithmetic sectors, while supplying general coefficient formulas at every layer.

A Coefficient tables

A.1 Carrier polynomials

For convenience, the next two carrier polynomials after those in [equation \(9.11\)](#) are

$$P_6(L) = -\frac{16}{15}L \left(20L^5 - 274L^4 + 1125L^3 - 1700L^2 + 900L - 120 \right), \quad (\text{A.1})$$

$$P_7(L) = \frac{32}{105}L \left(120L^6 - 2058L^5 + 11368L^4 - 25725L^3 + 24500L^2 - 8820L + 840 \right). \quad (\text{A.2})$$

They are generated without series guessing by [equation \(9.8\)](#).

A.2 Dominant inverse coefficients

The sequence b_m begins

$$1, \frac{5}{2}, \frac{13}{3}, \frac{53}{12}, -\frac{14}{5}, -\frac{763}{30}, -\frac{2179}{35}, -\frac{42289}{630}, \frac{132973}{1260}, \frac{3449711}{5040}, \dots \quad (\text{A.3})$$

A useful implementation check is the residual identity

$$U_M + \log(1 - z + zU_M) - 3 \log(1 + zU_M) = O(z^{M+1}), \quad (\text{A.4})$$

where $U_M = \sum_{m=1}^M b_m z^m$.

A.3 First logarithmic sector coefficients

For arbitrary k and σ , [equation \(5.6\)](#) gives

$$\ell_{k,\sigma,1} = -\frac{\sigma\lambda a}{2k} - \frac{\sigma k}{\lambda}, \quad (\text{A.5})$$

$$\ell_{k,\sigma,2} = a - \frac{k^2}{2\lambda^2}, \quad (\text{A.6})$$

$$\ell_{k,\sigma,3} = -\frac{\sigma\lambda a^2}{8k} - \frac{\sigma k a}{2\lambda} - \frac{\sigma k^3}{3\lambda^3}, \quad (\text{A.7})$$

$$\ell_{k,\sigma,4} = \frac{a^2}{2} - \frac{k^2 a}{2\lambda^2} - \frac{k^4}{4\lambda^4}. \quad (\text{A.8})$$

Consequently,

$$c_{k,\sigma,1} = \ell_{k,\sigma,1}, \quad (\text{A.9})$$

$$c_{k,\sigma,2} = \ell_{k,\sigma,2} + \frac{1}{2}\ell_{k,\sigma,1}^2, \quad (\text{A.10})$$

$$c_{k,\sigma,3} = \ell_{k,\sigma,3} + \ell_{k,\sigma,1}\ell_{k,\sigma,2} + \frac{1}{6}\ell_{k,\sigma,1}^3, \quad (\text{A.11})$$

$$c_{k,\sigma,4} = \ell_{k,\sigma,4} + \ell_{k,\sigma,1}\ell_{k,\sigma,3} + \frac{1}{2}\ell_{k,\sigma,1}^2\ell_{k,\sigma,2} + \frac{1}{2}\ell_{k,\sigma,1}^2\ell_{k,\sigma,2} + \frac{1}{24}\ell_{k,\sigma,1}^4. \quad (\text{A.12})$$

These are the first instances of [equation \(5.7\)](#).

B Derivation of the first inverse perturbation coefficients

For completeness, substitute $x = x_0 + \tau v_1 + \tau^2 v_2 + \tau^3 v_3 + \dots$ into

$$\Phi(x) + \tau\Psi(x) = R, \quad \Phi(x_0) = R.$$

At orders τ, τ^2, τ^3 one obtains

$$\begin{aligned} Dv_1 + \Psi &= 0, \\ Dv_2 + \frac{1}{2}\Phi''v_1^2 + \Psi'v_1 &= 0, \\ Dv_3 + \Phi''v_1v_2 + \frac{1}{6}\Phi'''v_1^3 + \Psi'v_2 + \frac{1}{2}\Psi''v_1^2 &= 0. \end{aligned}$$

Solving successively yields [equations \(11.31\) to \(11.33\)](#). The same calculation at arbitrary order is encoded compactly by partial Bell polynomials in [equation \(11.27\)](#).

C Normalization cross-checks

Several equivalent expressions are useful for detecting missing factors of k or $\sqrt{k}$. Starting from [equation \(3.1\)](#), the k th derivative factor is

$$\frac{d}{dn} \left(N^{-1/2} \sinh(x/k) \right) = \frac{1}{2N^{3/2}} \left(\frac{x}{k} \cosh(x/k) - \sinh(x/k) \right) \quad (C.1)$$

$$= \frac{x}{4kN^{3/2}} \left[e^{x/k} (1 - k/x) + e^{-x/k} (1 + k/x) \right]. \quad (C.2)$$

After multiplication by $\sqrt{k}/(\pi\sqrt{2})$, this may be written either as

$$\frac{1}{4\sqrt{3k}N} \left[e^{x/k} (1 - k/x) + e^{-x/k} (1 + k/x) \right] \quad (C.3)$$

or as

$$\frac{C}{\sqrt{k}x^2} \left[e^{x/k} (1 - k/x) + e^{-x/k} (1 + k/x) \right]. \quad (C.4)$$

For $k = 1$, the positive term is

$$\frac{e^x}{4\sqrt{3}N} \left(1 - \frac{1}{x} \right), \quad (C.5)$$

which tends to the Hardy–Ramanujan leading expression after replacing N by n and x by $\lambda\sqrt{n}$.

D Reference implementation blueprint

The following language-neutral pseudocode emphasizes exact coefficient arithmetic.

```
ForwardSectorCoefficients(k, sigma, M):
    ell[1..M] := formula (5.8)
    c[0] := 1
    for m = 1..M:
        c[m] := (1/m) * Sum_{j=1..m} j*ell[j]*c[m-j]
    return c

DominantInverseCoefficients(M):
    U := 0
    for m = 1..M:
        b[m] := -Coeff(z^m,
            log(1-z+z*U) - 3*log(1+z*U))
        U := U + b[m]*z^m
    return b

ArithmeticInverseCoefficients(K, residue, index_set):
    construct A_k(residue), gaps delta_j, amplitudes g_j
    v := 0
    for alpha in chosen monomial order over index_set:
        rhs := Sum_{m>=2} kappa_m*v^m
            + log(1 + Sum_j q_j*g_j(x0+v)*exp(-delta_j*v))
        V[alpha] := -Coeff(q^alpha, rhs)/D(x0)
        v := v + V[alpha]*q^alpha
    return V
```

For numerical work, evaluate logarithms of the dominant terms to avoid overflow, retain exact rational values for Dedekind sums as long as possible, and use the residual certificate of [theorem 13.5](#) after substituting numerical exponentials.

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